

Transistors and Optoelectronics

ELEC 2240

Chapter 1

v.13 (2020/2021)

Band Theory: Revision

All electronic devices rely on the flow of electrons under an electric potential/field.

The best way to understand electronic devices such as diodes and transistors is to look at the electron energetics.

If you connect a battery across a device, you give the electrons energy. If they then have sufficient energy to flow, you get a current. If it is energetically unfavourable for them to flow, get no current.



Get H&H fig 4.8 in here

幻灯片 2

W1

This is a natural waterscape picture. A stream flows over rocks, with surrounding vegetation. The white below mimics water reflection, creating a peaceful, natural vibe.

WPS_1582804925, 2025/7/18

Band Theory: Revision (2)

(If you can change between the two, you have a switch like a transistor).

Band theory is the study of electron energies in devices made of metals and semiconductors.



幻灯片 3

W2

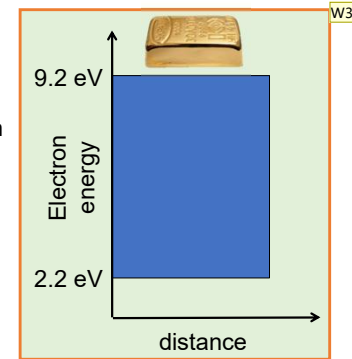
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Metal band

A 'band' is a graph of electron energy vs distance into the material.

It tells us what energies an electron can have in that material at a given location.



W3

This image shows an energy - band diagram related to a of a gold bar. Here's the breakdown:
Symbols and Terms Explanation

Electron energy: This vertical axis represents the energy levels that electrons in the material can occupy.

Distance: The horizontal axis represents the spatial dimension, showing how the electron energy levels vary with position in the material (though a simplified way here, as it might be a one - dimensional representation for basic understanding).

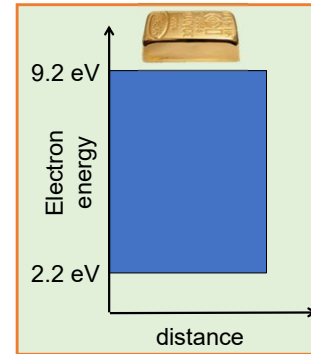
2.2 eV and 9.2 eV: These are energy values. In the context of energy bands, they could represent the boundaries of different electron energy regions. The range between 2.2 eV and 9.2 eV (a difference of 7 eV) might represent something like the energy range where electrons can have certain behaviors in this material. But for gold, which is a metal, the actual band structure is different from this simple representation. Pure gold has a very small band gap (more like a overlap of bands in reality), so this might be a simplified or hypothetical diagram for teaching basic energy - band concepts

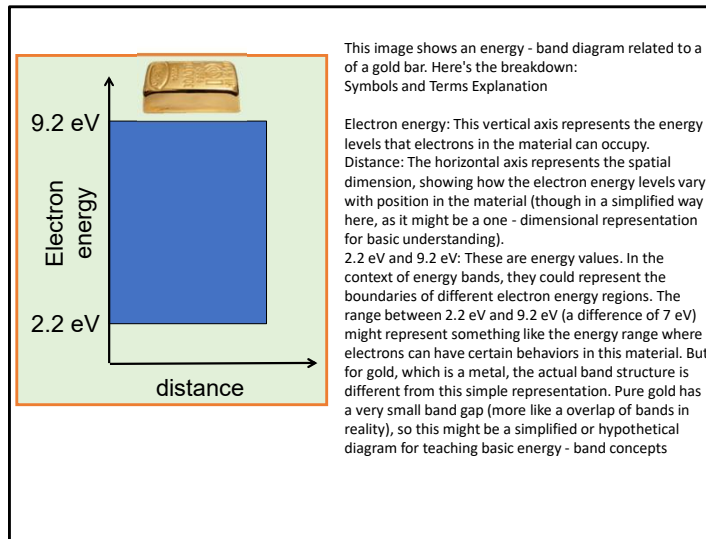
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Metal band (2)

Here, electrons can have in principle any energy between 2.2 eV and 9.2 eV.

But this graph doesn't show the electrons yet...

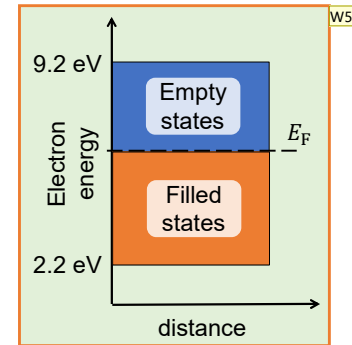




Metal band: occupancy

...when we include the electrons that are present in the material, they fill up the band from the bottom, like pouring water into a glass.

The lower energy states are full of electrons, whilst the upper energy states remain empty.



W5

This is an energy band diagram, likely for a metal (given the context of the previous gold - related diagram).

Electron energy: The vertical axis represents the energy levels that electrons in the material can have. The unit "eV" (electron volt) is a common unit for measuring the energy of electrons.

Distance: The horizontal axis represents the spatial dimension in a simplified way, showing the variation of electron energy levels with position in the material (in a very basic sense for introductory understanding).

2.2 eV and 9.2 eV: These are the boundary energy values of an energy band. The range between 2.2 eV and 9.2 eV represents an energy band where electrons can exist in this material.

EF(Fermi energy level): It is a horizontal dashed line. The Fermi energy level is the highest energy level that an electron can occupy at absolute zero temperature in a material. In this diagram, it divides the energy band into two parts.

Filled states: The orange - colored region with the label "Filled states" represents the energy levels within the band that are occupied by electrons. Electrons fill these states up to the Fermi energy level at a given temperature (simplified here).

Empty states: The blue - colored region with the label "Empty states" represents the energy levels within the band that are not occupied by electrons above the Fermi energy level.

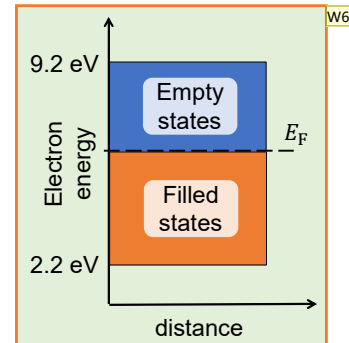
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Metal band: occupancy (2)

The highest occupied energy is called the **Fermi energy (E_F)**.

Materials which have a partly filled band like this are metals.

A metal with lots of electrons has a high E_F ; a metal with few electrons has a low E_F .



W6

This is an energy band diagram for a metal.

Electron energy: The vertical axis represents the energy levels that electrons in the material can have. The unit "eV" (electron volt) is a common unit for measuring the energy of electrons.

Distance: The horizontal axis represents the spatial dimension in a simplified way, showing the variation of electron energy levels with position in the material.

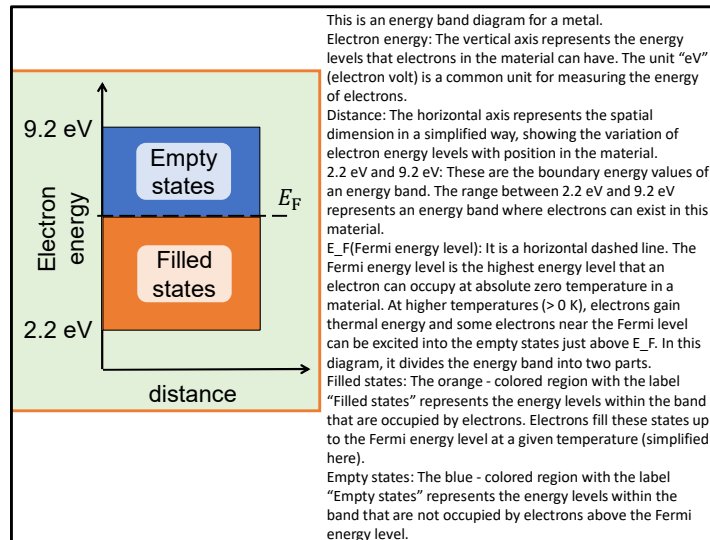
2.2 eV and 9.2 eV: These are the boundary energy values of an energy band. The range between 2.2 eV and 9.2 eV represents an energy band where electrons can exist in this material.

E_F (Fermi energy level): It is a horizontal dashed line. The Fermi energy level is the highest energy level that an electron can occupy at absolute zero temperature in a material. At higher temperatures (> 0 K), electrons gain thermal energy and some electrons near the Fermi level can be excited into the empty states just above E_F . In this diagram, it divides the energy band into two parts.

Filled states: The orange - colored region with the label "Filled states" represents the energy levels within the band that are occupied by electrons. Electrons fill these states up to the Fermi energy level at a given temperature (simplified here).

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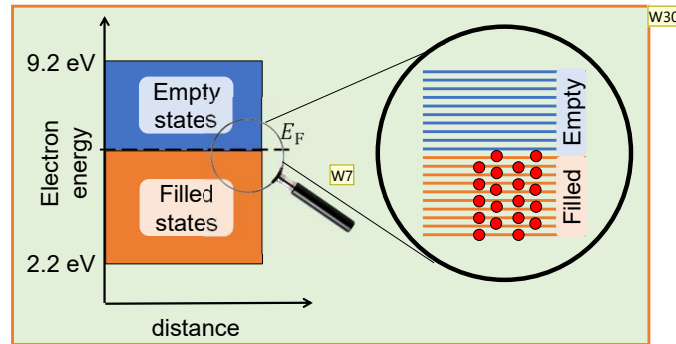


Metal band: state filling

Why do they fill up from the bottom?

The band is a series of closely spaced energy levels (energy 'states'), each of which can only hold two electrons.

Metal band: state filling (2)



W7

Electron energy (vertical axis): Measures energy levels electrons can occupy, in eV (a unit for tiny energy changes, like electron movement).

Distance (horizontal axis): Simplifies spatial layout, showing how electron energy relates to position in the material.

2.2 eV & 9.2 eV: Boundaries of an energy band—a range where electrons can exist. Think of it as a “lane” for electrons.

E

F

(Fermi energy level): The dashed line splitting the band. At very low temps, it's the highest energy level electrons fill.

Filled states (orange): Energy levels packed with electrons (below EF).

Empty states (blue): Energy levels with no electrons (above EF).

Magnified view: Zooms in to show energy levels as “lines” and electrons as red dots. Filled states have dots; empty ones don't.

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W30

Electron energy: The vertical axis represents the energy levels available to electrons, measured in electron volts (eV). It shows the range of energies that electrons can have within the material.

Distance: The horizontal axis represents the spatial dimension within the material, indicating how the electron energy levels are distributed as we move through the material.

Empty states (blue region): These are energy levels within the material that do not have electrons occupying them. Electrons can potentially move into these states if they gain enough energy.

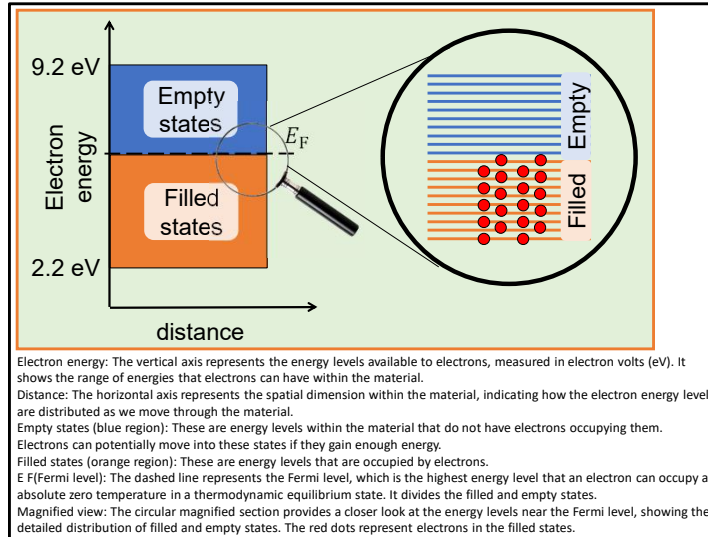
Filled states (orange region): These are energy levels that are occupied by electrons.

E F

(Fermi level): The dashed line represents the Fermi level, which is the highest energy level that an electron can occupy at absolute zero temperature in a thermodynamic equilibrium state. It divides the filled and empty states.

Magnified view: The circular magnified section provides a closer look at the energy levels near the Fermi level, showing the detailed distribution of filled and empty states. The red dots represent electrons in the filled states.

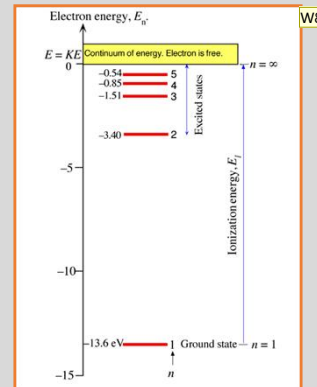
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Optional: Energy-band formation

If you are interested in knowing how these energy bands form, see Kasap Electronic Materials & Devices, Chapter 4.

Hydrogen has one electron – usually sits in ground state.



W8

This is an energy - level diagram of a hydrogen atom (or a hydrogen - like atom).

Electron energy (E_n) (vertical axis): Shows the energy of an electron in different states, in electron volts (eV). Negative values mean the electron is "bound" to the nucleus.

n (principal quantum number): Labels energy levels.

$n=1$ is the lowest (ground state);

$n=2,3,\dots$ are higher (excited states);

$n=\infty$ means the electron is free (ionized).

Ground state ($n=1$): The lowest energy level (-13.6eV). Electrons here are most tightly bound.

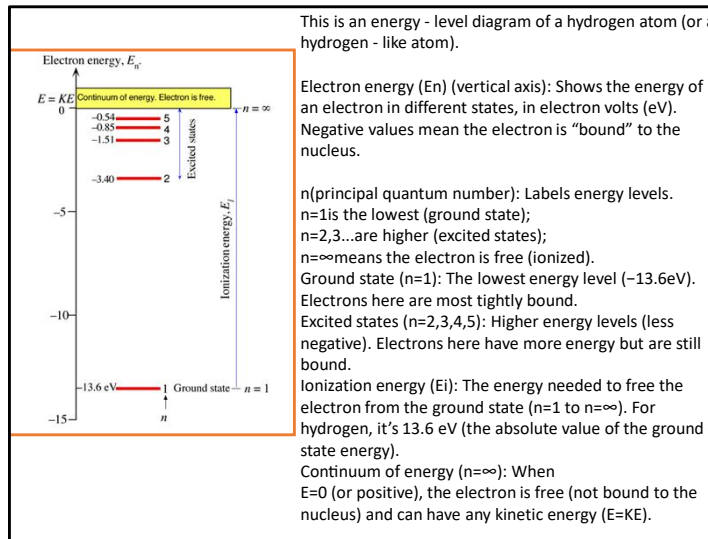
Excited states ($n=2,3,4,5$): Higher energy levels (less negative). Electrons here have more energy but are still bound.

Ionization energy (E_i): The energy needed to free the electron from the ground state ($n=1$ to $n=\infty$). For hydrogen, it's 13.6 eV (the absolute value of the ground - state energy).

Continuum of energy ($n=\infty$): When

$E=0$ (or positive), the electron is free (not bound to the nucleus) and can have any kinetic energy ($E=KE$).

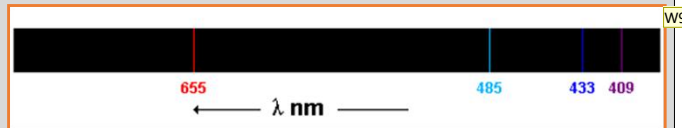
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Optional: Energy-band formation (2)

Emission spectrum indicates that atoms have quantized electron energy levels.

What happens if there is more than one atom, e.g. a crystal of metal?



W9

This is an atomic emission spectrum, likely from a specific element (e.g., hydrogen or a gas). Here's the breakdown:

Black background: Represents a range of wavelengths with no light emission.

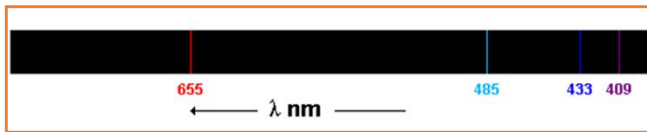
Colored lines: Show specific wavelengths (λ , in nanometers, nm) where the element emits light. Each color/wavelength corresponds to a unique electron transition in the atom.

Red (655 nm): Longest wavelength (lowest energy) emission.

Cyan (485 nm), Violet - blue (433 nm), Violet (409 nm): Shorter wavelengths (higher energy) emissions.

λ (wavelength): The distance between wave crests, measured in nanometers.

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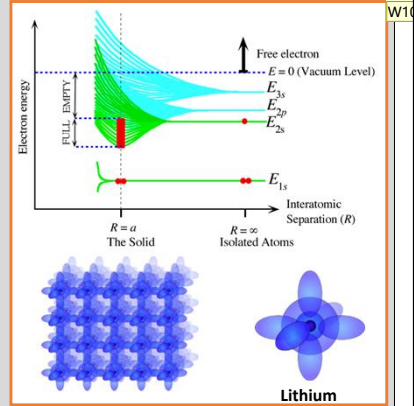
Cyan (485 nm), Violet - blue (433 nm), Violet (409 nm): Shorter wavelengths (higher energy) emissions.

λ (wavelength): The distance between wave crests, measured in nanometers.

Optional: Energy-band formation (3)

An atom has a series of discrete, separated energy levels.

As the atoms coalesce into a crystal, these energy levels split into very many closely spaced levels.



W10

This diagram explains how electron energy levels change when lithium atoms form a solid (like a lithium metal).

Electron energy (vertical axis): Shows energy levels of electrons.

Interatomic Separation (R) (horizontal axis): Distance between lithium atoms.

$R=\infty$: Isolated atoms (no interaction).

$R=a$: Atoms form a solid (close together, interacting).

Energy Levels (E 1s ,E 2s ,E 2p ,E 3s):

For isolated atoms ($R=\infty$), electrons occupy discrete levels (e.g., E 1s is filled, E 2s has some electrons).

For the solid ($R=a$), these levels split into many close - together levels (called a "band") due to atom - atom interactions.

FULL/EMPTY Band: The green/blue fan - shaped area shows a band of energy levels. "FULL" means levels are filled with electrons; "EMPTY" means they're not.

Vacuum Level (E=0): Energy of a free electron (not bound to any atom).

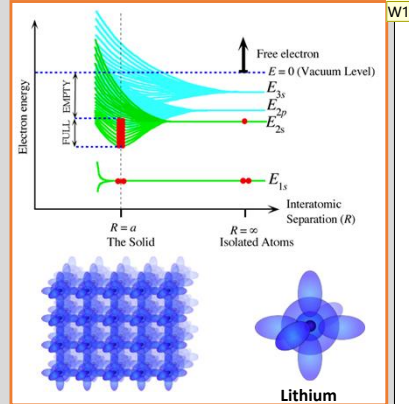
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Optional: Energy-band formation (4)

Each level can hold two electrons.

But they are so close together, they appear to form a continuum – an energy band.

Electrons minimise energy; populate band from the bottom.



W11

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Energy Levels (E 1s ,E 2s ,E 2p ,E 3s):

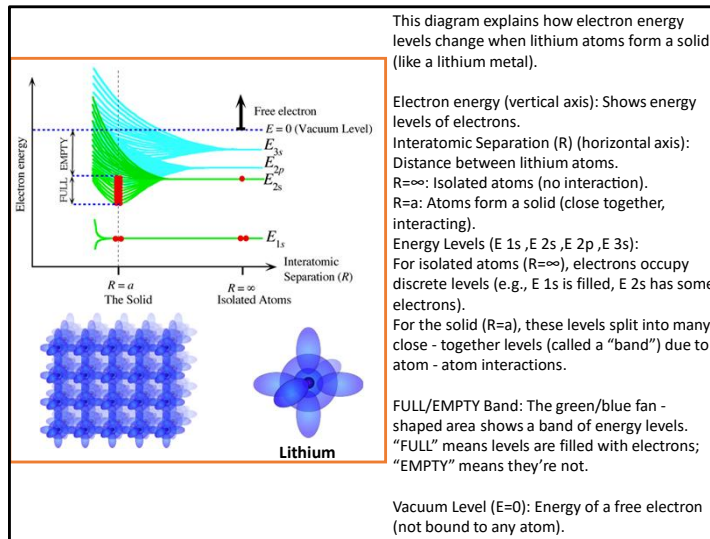
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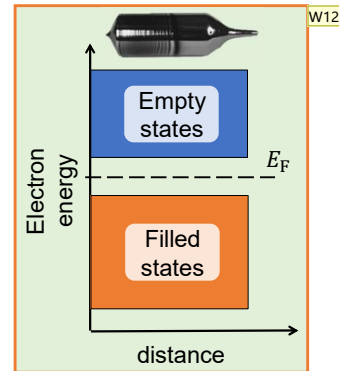
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Semiconductor band

A semiconductor has a different band structure to a metal.

It comprises two energy bands; the lower one is completely full, the upper one is completely empty (*undoped, and at low temperature*).



W12

Electron energy: The vertical axis represents the energy levels that electrons in a material can occupy. It helps us understand the energy distribution of electrons.

Distance: The horizontal axis shows the spatial dimension in a simplified way. It indicates how the electron energy levels change with position within the material, though this is a basic representation.

Filled states (orange region): This area contains energy levels that are occupied by electrons. Electrons fill these states up to a certain energy level at a given temperature.

Empty states (blue region): These are energy levels within the material that do not have electrons occupying them.

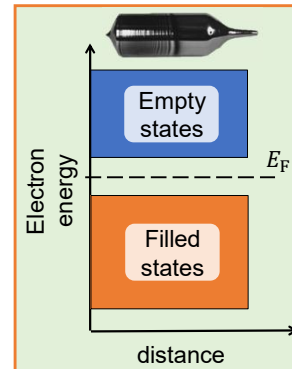
E F (Fermi energy level, dashed line): It is the highest energy level that an electron can occupy at absolute zero temperature in the material. It divides the energy levels into the filled and empty states. At temperatures above absolute zero, the distribution of electrons around this level determines the electrical properties of the material.

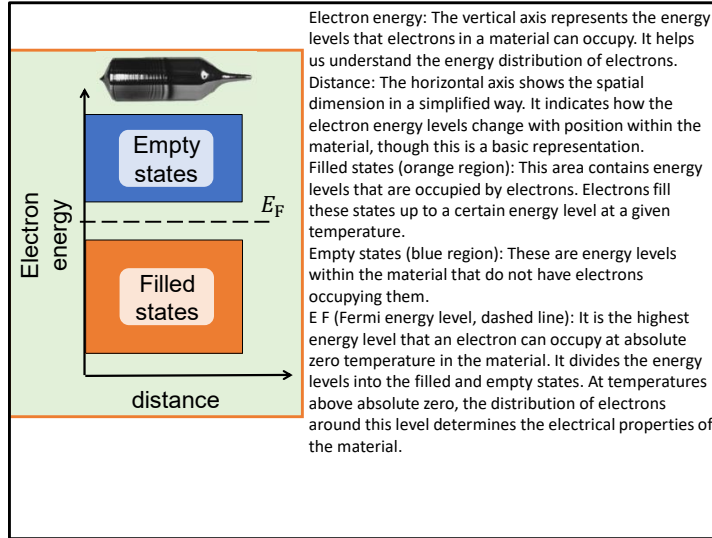
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Semiconductor band (2)

There is an **energy gap** E_G (**bandgap**) between the bands containing no energy states; electrons cannot be found here.

The Fermi energy lies approximately in the middle of the energy gap.



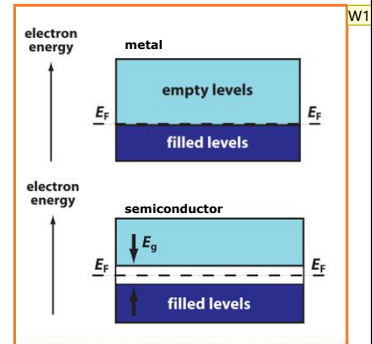


Metals vs semiconductors: recap

Two common situations:

If a band is part-full of electrons, this is a **metal**.

If one band is completely full (and is separated from next band), this is a **semiconductor**.



W13

This diagram compares the electron energy band structures of metals and semiconductors.

Electron energy: Vertical axis, representing energy levels electrons can occupy.

E_F(Fermi level): Dashed black line. It's the highest energy level electrons fill at absolute zero. Acts as a "reference" for electron distribution.

Filled levels: Dark blue region, where energy states are occupied by electrons.

Empty levels: Light blue region, where energy states are unoccupied (electrons can jump here).

E_g(Band gap, unique to semiconductors): Distance between arrows. It's the energy difference between the valence band (filled levels) and conduction band (empty levels). Electrons need at least

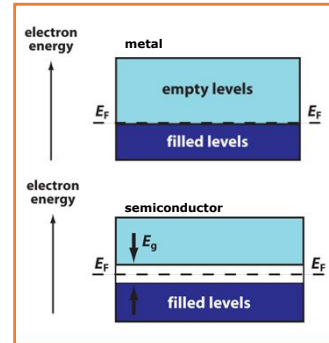
E_g energy to jump from the valence to the conduction band.

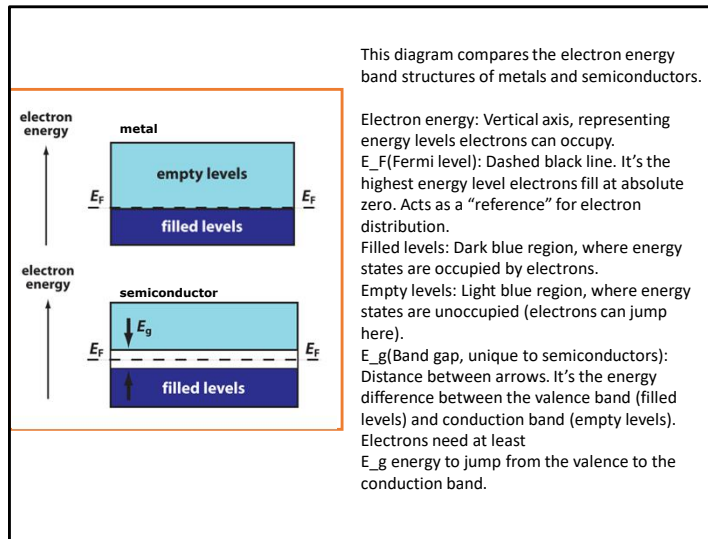
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Metals vs semiconductors: recap (2)

The distinction is whether the highest energy electrons have empty states (**metal**) or no states (**semiconductor**) above them.

We can define a Fermi energy (E_F) and, for the semiconductor, a band gap (E_G) where electrons are forbidden.

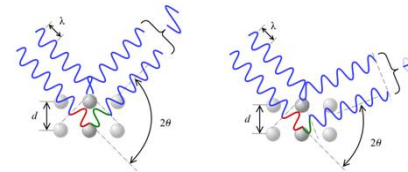




Bragg's Law

Bragg's Law: certain angles, wavelengths and lattice spacings lead to constructive and destructive interference of incident wave.

Bragg's Law:
 $2d \sin \theta = n\lambda$ for
 constructive
 interference.



W14

W13

W14

Spheres: They represent atoms in the crystal structure. Different - colored spheres indicate different types of atoms. This helps us visually distinguish various atomic species within the crystal.

"d": It stands for the inter - atomic distance, which is the distance between adjacent atoms in the crystal lattice. This distance is crucial as it affects the way X - rays interact with the crystal.

Blue wavy lines: These represent X - rays. X - rays are a form of electromagnetic radiation with short wavelengths that can penetrate through materials and interact with atoms in a crystal.

" λ ": It is the wavelength of the X - rays. The wavelength of X - rays is a key parameter in X - ray diffraction experiments. Different wavelengths can lead to different diffraction patterns.

" 2θ ": This is the diffraction angle. When X - rays interact with the crystal lattice, they are scattered at an angle. The diffraction angle is related to the inter - atomic distance "d" and the wavelength " λ " through Bragg's law.

Polarization: In the right - hand side diagram, the polarization of X - rays after diffraction is shown. Polarization refers to the orientation of the electric field vector of the electromagnetic wave (X - ray in this case).

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W15

This picture consists of three parts, all related to Sir William Bragg:

1. Left & Middle: Portraits of William Bragg and His Son

Who they are:

The two men are Sir William Bragg (left) and his son Sir Lawrence Bragg (middle). They were a famous father - son duo in physics. Both won the Nobel Prize in Physics (1915) for their work on X - ray crystallography (using X - rays to study crystal structures).

2. Right: Commemorative Plaque

Text breakdown:

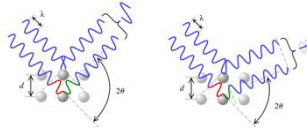
"SIR WILLIAM BRAGG (1862–1942)": His name and lifespan.

"Cavendish Professor of Physics at the University of Leeds – 1909–1915": His academic position (he taught physics at the University of Leeds).

"Together with his son Lawrence he perfected the X - ray analysis of crystal structures.": Highlights their Nobel - winning work—using X - rays to "see" and understand how atoms arrange in crystals.

"Joint Nobel Laureate for Physics in 1915 with his son Sir Lawrence Bragg": Confirms their shared Nobel Prize.

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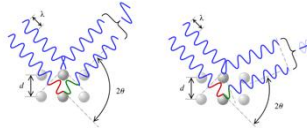
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Bragg's Law tells us how waves interact with a crystal lattice. Normally, we use it to explain X-ray diffraction, but the same principle applies to electron waves inside a solid.

Electrons in a crystal behave like waves. When these waves travel through the periodic arrangement of atoms, they can be reflected (diffracted) if the Bragg condition is satisfied. At certain special points in momentum space (the edges of the Brillouin zone), the electron waves interfere in such a way that they cannot propagate freely.

This "blocking" of electron waves creates an energy gap, which shows up in the band diagram as the forbidden region between the valence band and the conduction band.



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Highlights their Nobel - winning work—using X - rays to “see” and understand how atoms arrange in crystals.

“Joint Nobel Laureate for Physics in 1915 with his son Sir Lawrence Bragg”: Confirms their shared Nobel Prize.

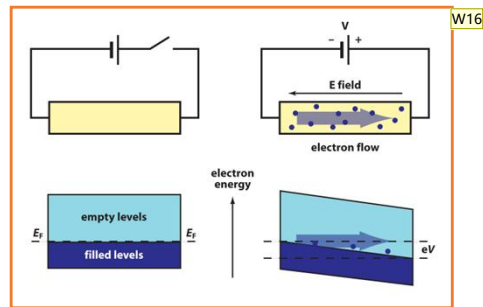
Conduction

If a bias is dropped across the material, the bands bend to reflect the change in electron energy across the material.

Here we assume the bias drops uniformly. Therefore, E_F drops uniformly by eV . (E_F is constant in a material in thermal equilibrium, and not subject to external excitation such as biases or illumination).

Conduction: metals

The energy band diagram shows a graph of electron energy vs distance. Bias causes the bands to tilt.



W16

This diagram explains how a voltage creates electric current in a metal by linking circuit behavior to electron energy levels.

Top Circuit Diagrams:

Battery/Voltage (

V

): Supplies energy to push electrons.

"E field" (Electric Field): Forces electrons to move (direction shown by the arrow).

"electron flow": Blue dots (electrons) moving in the metal.

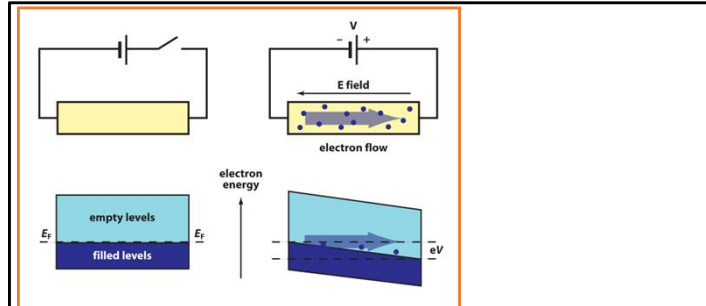
Bottom Energy Level Diagrams:

E F Fermi Level): Dashed line, the highest energy level electrons fill at 0K.

"filled levels": Blue region, energy states occupied by electrons.

"empty levels": Light blue region, energy states electrons can jump into.

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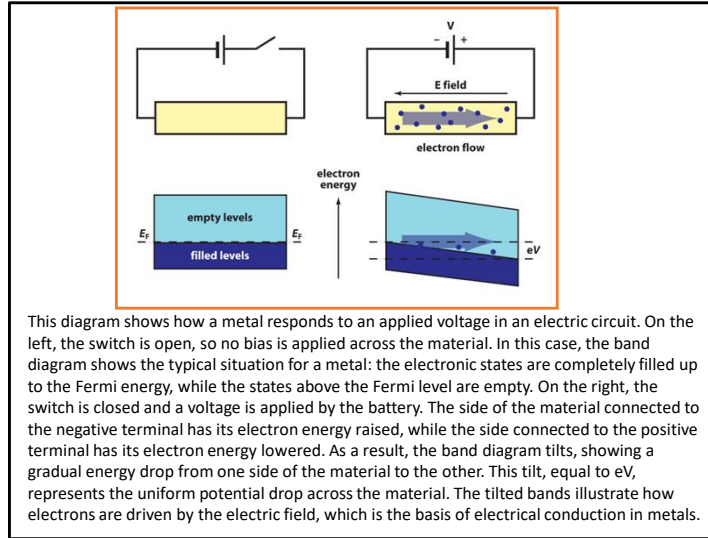
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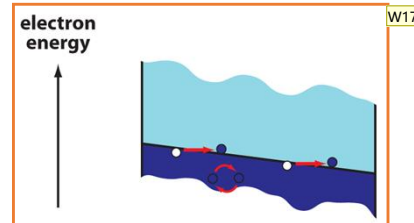
"filled levels": Blue region, energy states occupied by electrons.

"empty levels": Light blue region, energy states electrons can jump into.



Conduction: metals (2)

All conduction takes place by electrons at the Fermi energy; these are the only electrons which have adjacent empty states into which they can move ('drift') under influence of the electric field. **Think of water in a tube!**



W17

This diagram illustrates electron scattering in a metal or semiconductor—a key concept in understanding electrical resistance.

Electron energy (vertical axis): Shows energy levels available to electrons.

Blue (top): “Empty” energy states electrons can jump into.

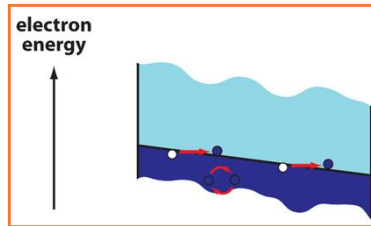
Dark blue (bottom): “Filled” energy states, with:

White circles: Atoms in the material’s lattice.

Blue circles: Electrons moving through the lattice.

Red circle/arrow: A defect or impurity causing electron scattering.

WPS_1582804925, 2025/9/1



This diagram illustrates electron scattering in a metal or semiconductor—a key concept in understanding electrical resistance.

Electron energy (vertical axis): Shows energy levels available to electrons.

Blue (top): “Empty” energy states electrons can jump into.

Dark blue (bottom): “Filled” energy states, with:

White circles: Atoms in the material’s lattice.

Blue circles: Electrons moving through the lattice.

Red circle/arrow: A defect or impurity causing electron scattering.

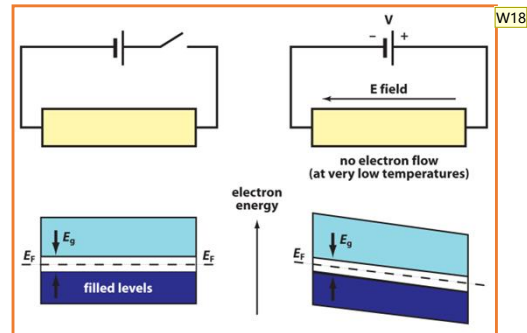
Conduction: metals (3)

The electrons deep down below the Fermi surface can swap positions, but the overall effect of this leaves the system unchanged.

This is known as a **degenerate electron system**, since most electrons might as well not be there (they are 'degenerate').

The physical properties of metals (e.g. electrical and thermal conduction) are therefore governed by just the electrons at the Fermi surface.

Conduction: semiconductors



W18

This diagram explains why semiconductors stop conducting at very low temperatures by linking circuit behavior to their energy band structure.

Top Circuit Diagrams:

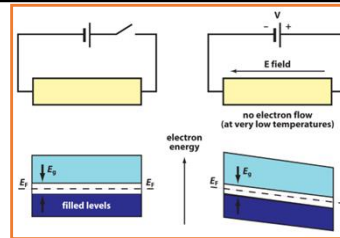
Battery/Voltage (V): Applies an electric field (E field) to the material.

"no electron flow (at very low temperatures)": At low temps, the semiconductor acts as an insulator.

Bottom Energy Level Diagrams:

E_g (Band gap): Energy needed for electrons to jump from filled levels (blue) to empty levels (light blue).

E_F (Fermi level): Dashed line, the highest energy level electrons fill at 0K.



This diagram shows the behavior of a semiconductor in an electric circuit. On the left, the switch is open, and no voltage is applied across the yellow block of material. The band diagram underneath represents a semiconductor at very low temperatures: the lower energy band (valence band) is completely filled with electrons, the upper energy band (conduction band) is completely empty, and the two are separated by an energy gap E_g where no electronic states exist. On the right, the switch is closed and a voltage is applied. The electrons on the left side of the semiconductor, near the negative terminal of the battery, have their energy raised, while the electrons on the right side, near the positive terminal, have their energy lowered. This produces a tilt in the band structure, reflecting the uniform potential drop across the material. However, unlike in metals, the presence of the band gap prevents conduction: electrons in the valence band cannot move into the conduction band because there are no available states in the band gap. As a result, at very low temperatures and under moderate electric fields, the semiconductor does not conduct current.

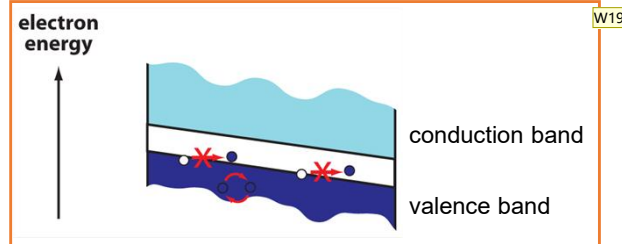
Conduction: semiconductors (2)

In a semiconductor, a forbidden energy gap E_G exists between the filled band (**valence band**) and the empty band (**conduction band**).

Under a bias V , the bands still bend, but now, unlike in a metal, there are no empty states for the highest energy electrons to scatter into to conduct through the material. Therefore, no conduction.

Conduction: semiconductors (3)

Because of the energy gap, no electrons can move under the influence of the applied electric field; very high resistance (at low temperatures).



W19

This diagram illustrates electron behavior in a semiconductor's energy bands, focusing on how electrons move (and get scattered) between the valence and conduction bands.

Electron energy (vertical axis): Shows energy levels available to electrons.

Valence band (dark blue): Lower energy band, where electrons are "bound" to atoms (filled with electrons at 0K).

Conduction band (light blue): Higher energy band, where electrons are free to move (empty at 0K in pure semiconductors).

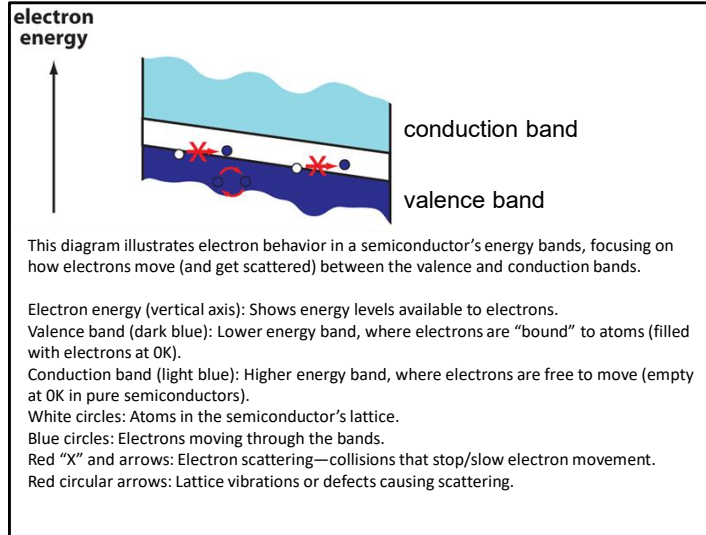
White circles: Atoms in the semiconductor's lattice.

Blue circles: Electrons moving through the bands.

Red "X" and arrows: Electron scattering—collisions that stop/slow electron movement.

Red circular arrows: Lattice vibrations or defects causing scattering.

WPS_1582804925, 2025/9/1



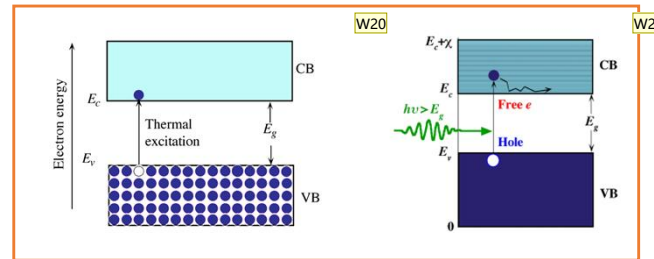
Conduction: semiconductors (4)

There are several ways of controllably making a semiconductor conduct though; this is why they are so important in electronics.

1. Excite electrons from valence band into conduction band (by light or heat);
2. Incorporate impurity (dopant) atoms into the pure semiconductor.

Electrons and holes

An electron can be promoted from the valence band to the conduction band by heat or by light – it leaves behind an unfilled state – a **hole**.



W20

This diagram illustrates thermal excitation in a semiconductor, explaining how heat can push electrons from the valence band to the conduction band.

Electron energy (vertical axis): Shows energy levels available to electrons.

VB (Valence Band, dark blue): Lower energy band, filled with electrons (dots) that are bound to atoms.

CB (Conduction Band, light blue): Higher energy band, where electrons can move freely (initially empty in pure semiconductors).

E_g (Band Gap): Energy difference between VB and CB—electrons need this energy to jump from VB to CB.

E_v (Valence Band Edge): Top energy level of the VB.

E_c (Conduction Band Edge): Bottom energy level of the CB.

"Thermal excitation": Heat provides energy to electrons, letting them jump across E_g

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W21

This diagram illustrates photon - induced electron - hole pair generation in a semiconductor, a key process in solar cells and photodetectors.

Electron energy (vertical axis): Shows energy levels available to electrons.

VB (Valence Band, dark blue): Lower energy band, filled with electrons (initially full, shown as a solid color).

CB (Conduction Band, light blue): Higher energy band, where electrons can move freely (initially empty).

E_g

(Band Gap): Energy difference between VB and CB—minimum energy needed for an electron to jump from VB to CB.

E_v

(Valence Band Edge): Top energy level of the VB.

E_c

(Conduction Band Edge): Bottom energy level of the CB.

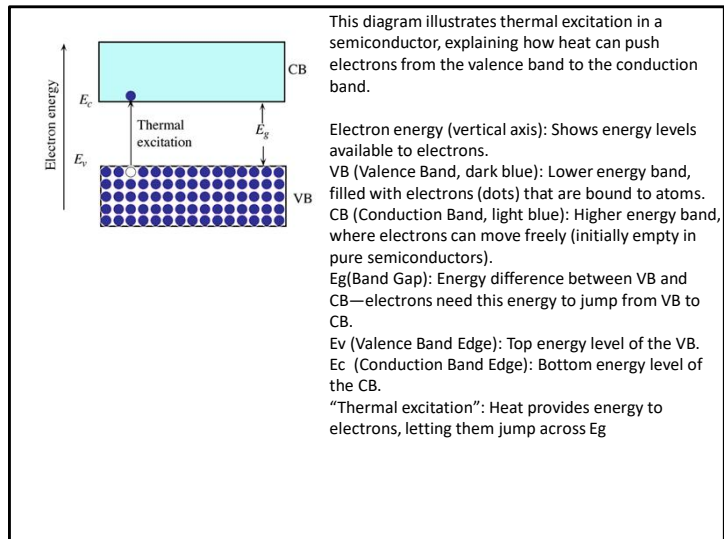
$h\nu$ (Photon Energy): Energy of incoming light (green wave). Requires

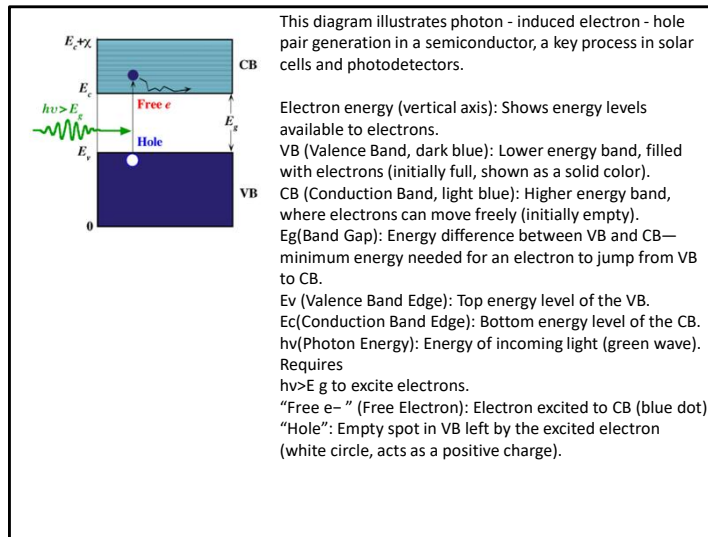
$h\nu > E_g$ to excite electrons.

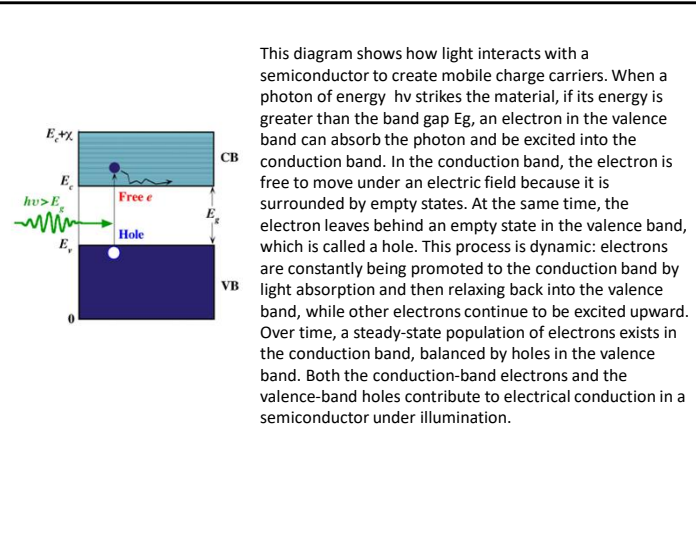
"Free e^- " (Free Electron): Electron excited to CB (blue dot).

"Hole": Empty spot in VB left by the excited electron (white circle, acts as a positive charge).

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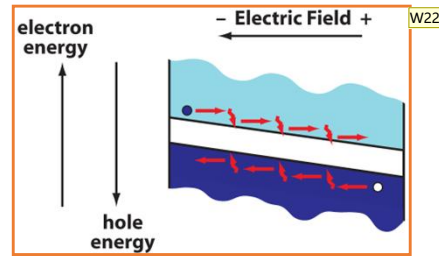
Electrons and holes (2)

Just an electron can conduct through a partly-filled conduction band; a hole can conduct through a partly-filled valence band.

It's dynamic; the electrons will drop back to the valence band after a while (**recombination**), but other electrons will be promoted.

Electrons and holes (3)

Under an electric field, the electron gains energy from the field and accelerates. It will scatter off an impurity or thermal vibration (phonon) and lose energy, falling down the band.



W22

This diagram shows how electrons and holes move in a semiconductor under an electric field, explaining charge transport.

Electron energy (left vertical axis): Tracks electron energy levels.

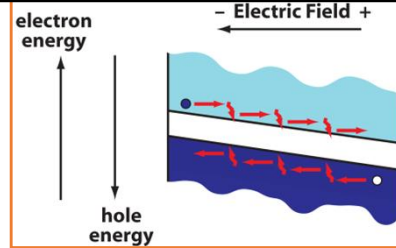
Hole energy (right vertical axis, inverted): Holes act like positive charges, so their energy increases downward.

Electric Field (+ to -): Pushes electrons left and holes right (opposite directions, since electrons are negative, holes are positive "defects").

CB (Conduction Band, light blue): Where free electrons (blue dot) move.

VB (Valence Band, dark blue): Where holes (white circle) move (holes are empty spots in the VB).

WPS_1582804925, 2025/9/1



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VB (Valence Band, dark blue): Where holes (white circle) move (holes are empty spots in the VB).

Electrons and holes (4)

After the collision, it picks up kinetic energy again from the field, accelerates, and the process repeats. The electron **drifts** through the crystal under the applied field at a (average) drift velocity v_{de} .

The hole drifts in the same way at a (average) drift velocity v_{dh} ; note holes move as if they have a **positive** charge.

Intrinsic and extrinsic semiconductors

Pure semiconductor crystals are **intrinsic** semiconductors. Since every conduction band electron was promoted from the valence band (leaving behind a hole), the number of electrons equals the number of holes, $n = p$.

However, the reason that semiconductor materials are the basis for a multi-billion-dollar industry is that they can be **doped**.

Intrinsic and extrinsic semiconductors (2)

Doping is the controlled introduction of small quantities of impurities into intrinsic semiconductors which locally adds extra electrons (or extra holes), allowing the conduction properties of the crystal to be altered locally.

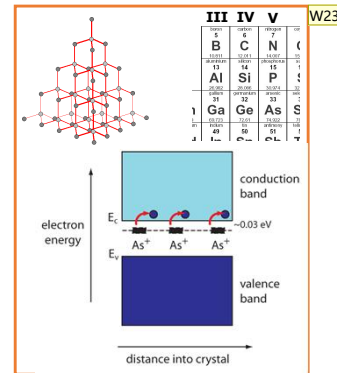
Doped semiconductors are called **extrinsic** semiconductors (since the electrons or holes are provided external to the main material).

***n*-type semiconductor**

If you dope silicon with arsenic:

The energy required to liberate the spare As electron into the silicon crystal is very small (~ 0.03 eV, c.f. $E_g = 1.1$ eV for silicon).

The As atoms form states just (0.03 eV) below the CB edge.



W23

This diagram explains n - type doping in a semiconductor (using arsenic to dope silicon), a key process in making electronic devices.

Periodic Table (Top Right): Shows groups III, IV, V. Silicon (Si, group IV) is the host semiconductor; arsenic (As, group V) is the dopant.

Crystal Structure (Top Left): 3D lattice of silicon atoms (gray) with covalent bonds (red). Doping adds arsenic atoms.

Energy Bands (Bottom):

Conduction Band (light blue): Where free electrons move.

Valence Band (dark blue): Filled with bound electrons in pure Si.

E_c

(Conduction Band Edge),

E_v

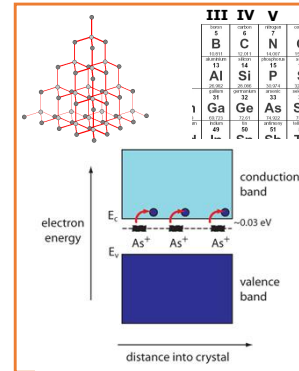
(Valence Band Edge): Energy boundaries of the bands.

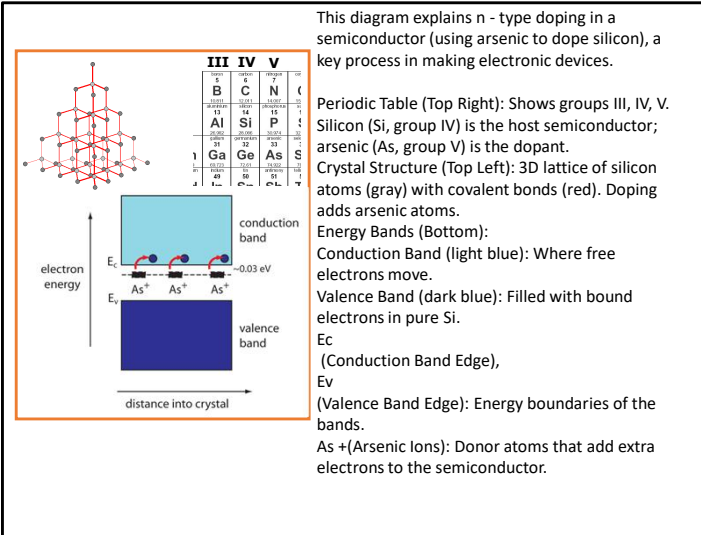
As^+ (Arsenic Ions): Donor atoms that add extra electrons to the semiconductor.

n-type semiconductor (2)

At low temperatures, the spare electrons are localized (trapped) on the As atoms. But, at room temperature, the thermal energy exceeds this binding energy six-fold, therefore the spare As electrons are liberated into the conduction band.

Arsenic is a **donor** for silicon since it donates electrons into the CB.



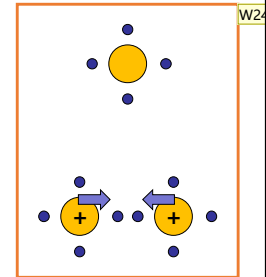


***n*-type semiconductor (3)**

Silicon has four outer electrons,
used for bonding.

Negative charge pulls the positive
nuclei together to form the chemical
bond.

These are the valence band
electrons and cannot conduct.



W24

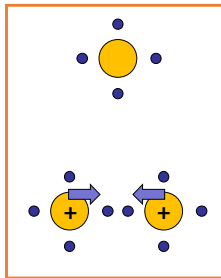
This diagram shows a simplified model of ion movement in a crystal lattice, likely illustrating how charged particles (ions) interact and migrate.

Large yellow circles: Positively charged ions (labeled "+")—the main atoms/ions in the lattice.

Small purple circles: Other ions/atoms in the crystal structure (could be neutral or charged, context - dependent).

Arrows: Show direction of ion movement (yellow ions shift positions).

WPS_1582804925, 2025/9/1



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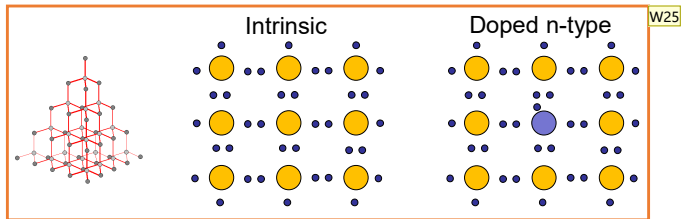
Small purple circles: Other ions/atoms in the crystal structure (could be neutral or charged, context - dependent).

Arrows: Show direction of ion movement (yellow ions shift positions).

***n*-type semiconductor (4)**

The silicon atoms are bonded into a diamond-lattice formation.

The arsenic atoms do not change the overall lattice structure, but they add extra free electrons to the conduction band.



W25

This diagram compares intrinsic and n - type doped semiconductors, showing how doping changes a material's atomic structure.

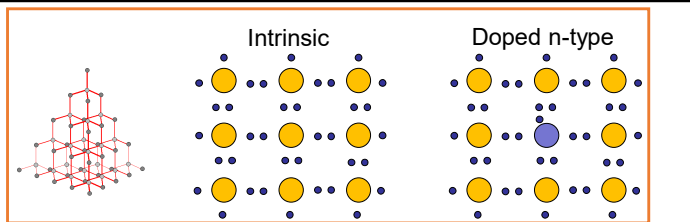
3D Lattice (Left): Represents the crystal structure of a semiconductor (e.g., silicon), with atoms connected by covalent bonds.

Yellow circles: Host atoms (e.g., silicon, group IV in the periodic table).

Small purple circles: Valence electrons (4 per yellow atom in intrinsic semiconductors).

Large purple circle (Doped n - type): Dopant atom (e.g., arsenic, group V), which has 5 valence electrons.

WPS_1582804925, 2025/9/1



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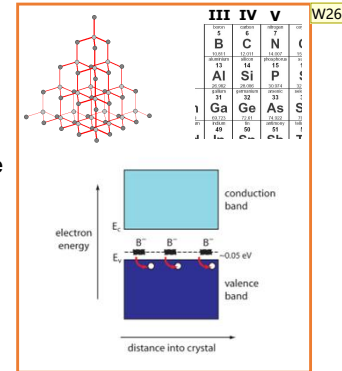
Large purple circle (Doped n - type): Dopant atom (e.g., arsenic, group V), which has 5 valence electrons.

p-type semiconductor

If you dope silicon with boron:

As before, the energy required to liberate the B hole (or equivalently, to capture a silicon valence electron) is very small (~ 0.05 eV, c.f. $E_g = 1.1$ eV for silicon).

The B atoms form states just above the VB edge.



W26

This diagram explains p - type doping in a semiconductor (using boron to dope silicon), a fundamental process in semiconductor device fabrication.

Periodic Table (Top Right): Displays groups III, IV, and V. Silicon (Si, group IV) is the host semiconductor, and boron (B, group III) is the dopant.

Crystal Structure (Top Left): Shows the 3D lattice of silicon atoms (gray) connected by covalent bonds (red). Doping introduces boron atoms into this lattice.

Energy Bands (Bottom):

Conduction Band (Light Blue): Region where free electrons can move; initially empty in pure silicon for p - type doping context.

Valence Band (Dark Blue): Region filled with bound electrons in pure silicon; doping creates vacancies (holes) here.

E_c

(Conduction Band Edge): The lowest energy level of the conduction band.

E_v (Valence Band Edge): The highest energy level of the valence band.

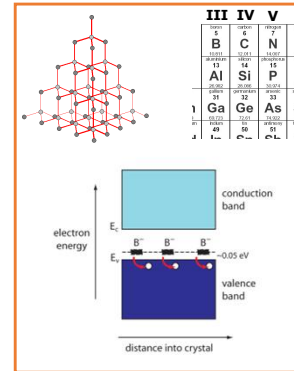
B^- (Boron Ions): Acceptor atoms that create holes in the valence band.

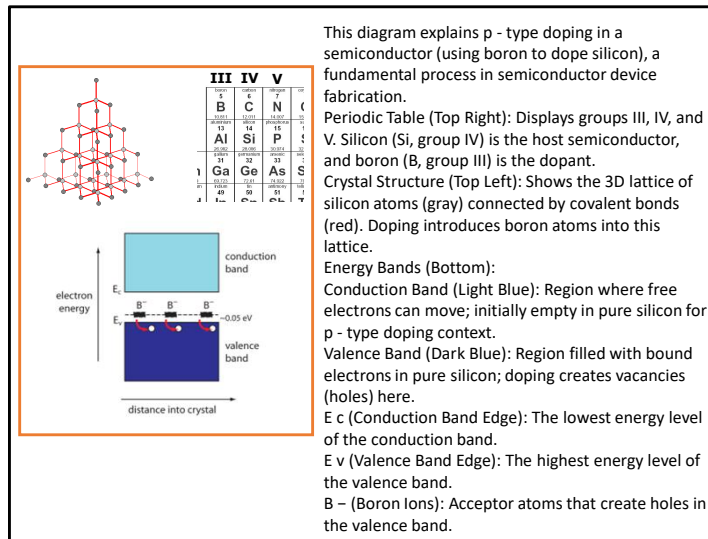
WPS_1582804925, 2025/9/1

p-type semiconductor (2)

At room temperature, the thermal energy allows valence electrons to hop onto the B atoms (making them B ions), thus liberating holes into the silicon valence band which can conduct.

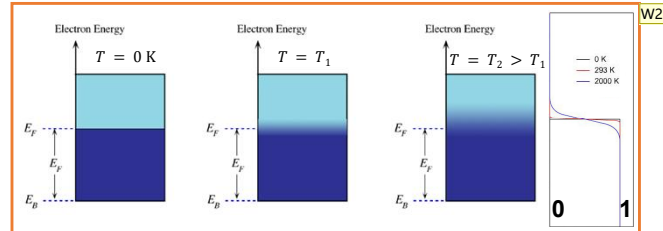
Boron is an **acceptor** for silicon since it accepts electrons from the VB (liberating holes).





Fermi-Dirac function $f(E)$

The previous diagrams have been at zero Kelvin (absolute zero). As temperature increases, electrons gain thermal energy and jump into empty energy levels above the original Fermi energy.



W27

This diagram explains how electron energy distribution in a material changes with temperature, using the Fermi - Dirac distribution.

Electron Energy (vertical axis): Shows energy levels available to electrons.

E_F

(Fermi Level): Dashed line, the highest energy level electrons fill at 0K.

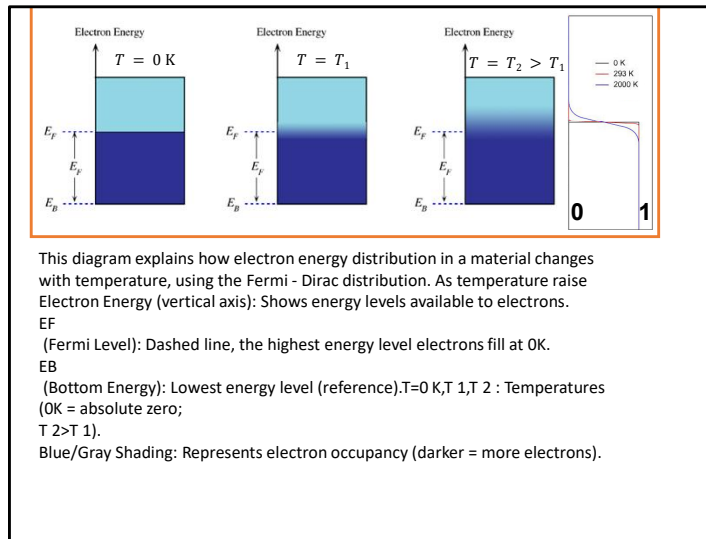
E_B

(Bottom Energy): Lowest energy level (reference). $T=0\text{ K}$, T_1 , T_2 : Temperatures (0K = absolute zero;

$T_2 > T_1$).

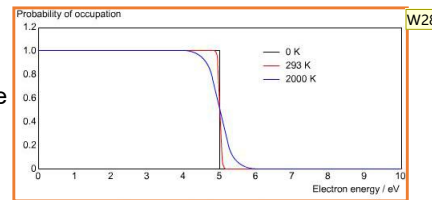
Blue/Gray Shading: Represents electron occupancy (darker = more electrons).

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Fermi-Dirac function $f(E)$ (2)

The Fermi-Dirac function $f(E)$ tells you the probability that an energy state at energy E is occupied, at any given temperature.



$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_b T}\right)}$$

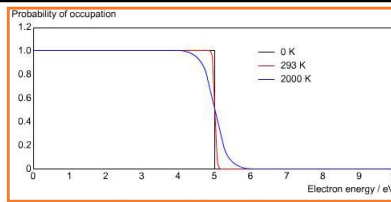
W28

Probability of occupation: The vertical axis represents the likelihood that an electron energy level is occupied by an electron. A value of 1 means the level is always occupied, and 0 means it is never occupied.

Electron energy (eV): The horizontal axis shows the energy of electron levels in electron volts. This helps us understand at which energy levels electrons are likely to be found.

0 K, 293 K, 20000 K: These are different temperatures. 0 K is absolute zero (theoretical lowest temperature), 293 K is approximately room temperature, and 20000 K is a very high temperature. The curves for different temperatures show how electron occupancy changes with temperature.

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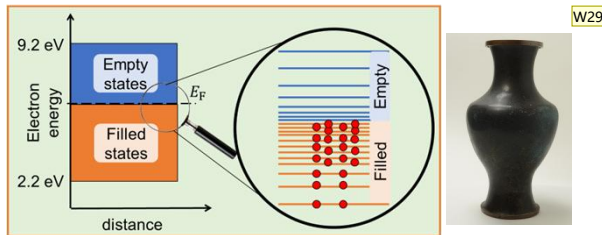
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Density of States $g(E)$

The energy states are not equally spaced; a quantity called the 'density of states' $g(E)$ tells you how many states you have at energy E (per volume).



Can note the DoS goes to zero at bottom, prob for lasers. Explains why different metals conduct differently.

W29

Electron energy: The vertical axis represents the energy levels available to electrons, measured in electron volts (eV). It shows the range of energies that electrons can have within the material.

Distance: The horizontal axis represents the spatial dimension within the material, indicating how the electron energy levels are distributed as we move through the material.

Empty states (blue region): These are energy levels within the material that do not have electrons occupying them. Electrons can potentially move into these states if they gain enough energy.

Filled states (orange region): These are energy levels that are occupied by electrons.

E F

(Fermi level): The dashed line represents the Fermi level, which is the highest energy level that an electron can occupy at absolute zero temperature in a thermodynamic equilibrium state. It divides the filled and empty states.

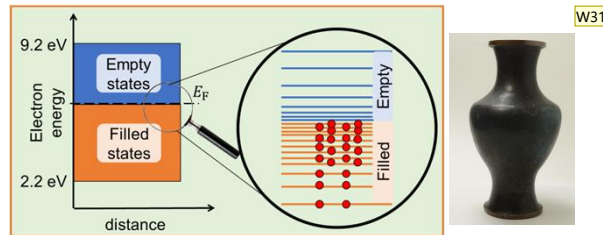
Magnified view: The circular magnified section provides a closer look at the energy levels near the Fermi level, showing the detailed distribution of filled and empty states. The red dots represent electrons in the filled states.

WPS_1582804925, 2025/7/18

Density of States $g(E)$ (2)

In 3D materials, $g(E)$ is largest at the centre of the band:

$$g_{3D}(E) = \frac{m_0}{h^3 \pi^2} \sqrt{2m_0 E}$$



W31

Electron energy: The vertical axis represents the energy levels available to electrons, measured in electron volts (eV). It shows the range of energies that electrons can have within the material.

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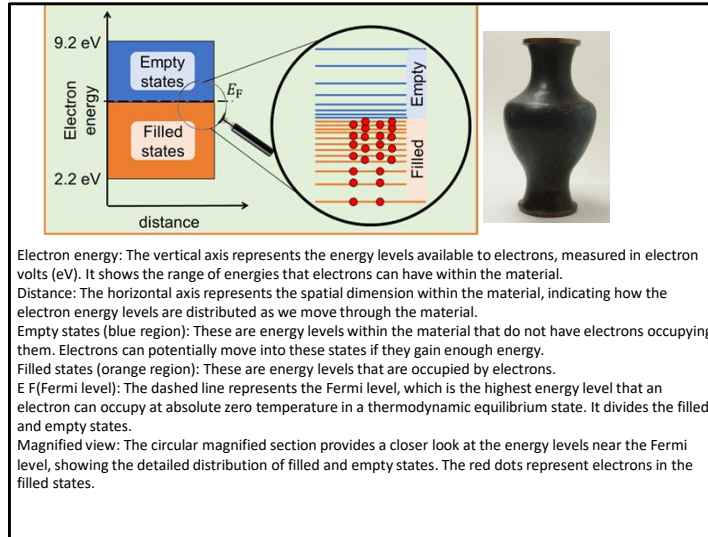
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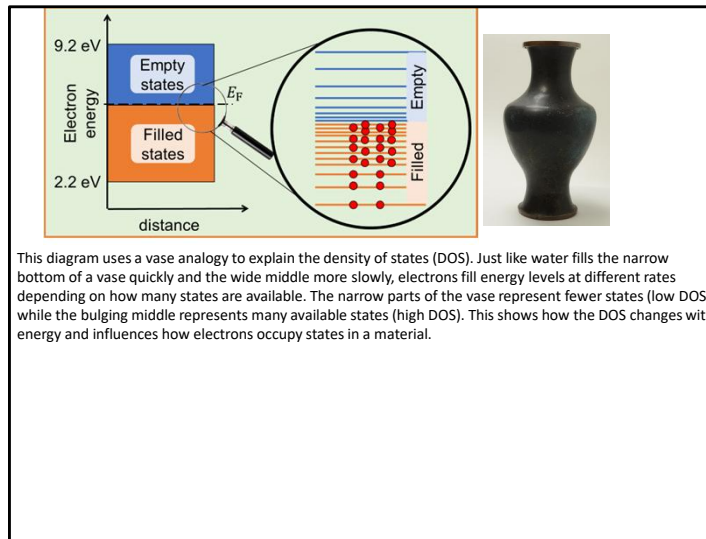
E F

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Magnified view: The circular magnified section provides a closer look at the energy levels near the Fermi level, showing the detailed distribution of filled and empty states. The red dots represent electrons in the filled states.

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Number of electrons in band

We can use the density of states $g(E)$ and the Fermi-Dirac function $f(E)$ to calculate how many electrons there are in a band at arbitrary temperature.

- $g(E)$ tells us how many states exist around a given energy E ;
- $f(E)$ tells us the probability that such a state will be occupied.

Therefore, total number of electrons in band:

$$\text{number} = \int_{\text{band}} f(E)g(E) \, dE$$

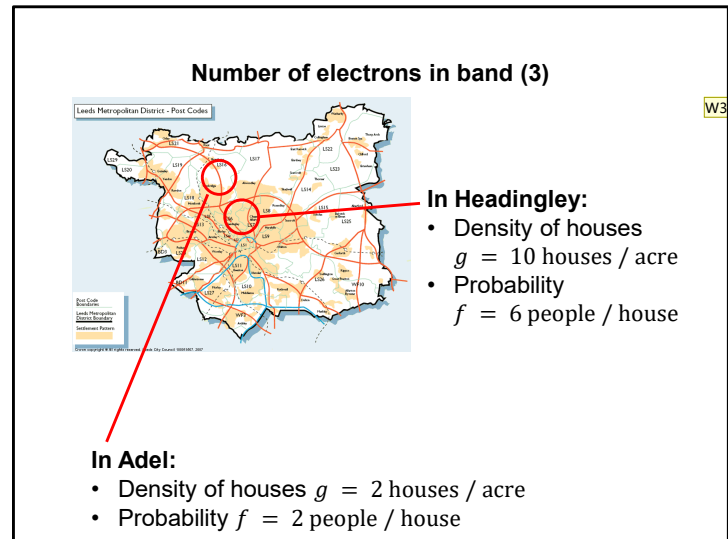
Number of electrons in band (2)

Use an analogy to understand the formula:

$$\text{number} = \int_{\text{band}} f(E)g(E) \, dE$$

Suppose you were counting all the people living in Leeds...

Instead of counting every house one-by-one, work out the average population per house, and the average house density per suburb and then add up (integrate) over the city.



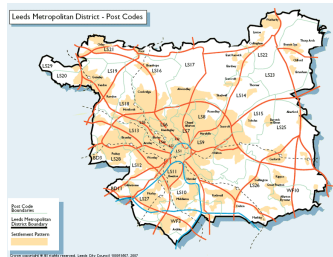
Bit like census, so how many houses in an area and how likely they are to be filled i.e. how many people in them. So in inner city, lots of closely packed houses, or blocks of flats, so DoS is high, but go into the country, houses are fewer so DoS is lower.

W33

This image presents housing and population data for two areas (Headingley and Adel) in the Leeds Metropolitan District, using a map and statistics.

Leeds Metropolitan District Map: Shows postcode areas, with red circles highlighting Headingley and Adel.
Red Lines: Connect the map regions to their respective data (Headingley top, Adel bottom).

WPS_1582804925, 2025/9/1



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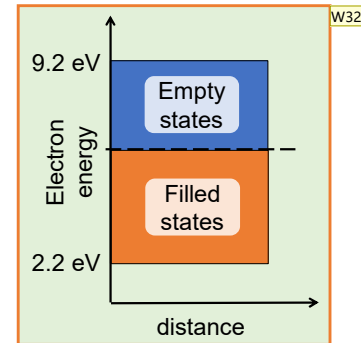
Number of electrons in band (4)

$$n = \int_{\text{band}} f(E)g(E) dE$$

Where:

$$g_{3D}(E) = \frac{m_0}{\hbar^3 \pi^2} \sqrt{2m_0 E}$$

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_b T}\right)}$$



Number of electrons in band (5)

At $T = 0 \text{ K}$, $f(E) = 1$ or 0 . The integral is zero for $E >$ Fermi energy.

$$\begin{aligned} n &= \int_0^{E_F} g(E) \, dE \\ &= 8\pi\sqrt{2} \left(\frac{m_0}{h^2}\right)^{\frac{3}{2}} \int_0^{E_F} E^{1/2} \, dE = 8\pi\sqrt{2} \left(\frac{m_0}{h^2}\right)^{\frac{3}{2}} \frac{2}{3} E_F^{3/2} \end{aligned}$$

$$E_F = \frac{h^2}{8m_0} \left(\frac{3n}{\pi}\right)^{2/3}$$

Formally relates the Fermi energy to number of electrons in band.

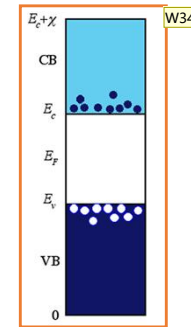
Electron and hole concentration

How do we determine electron and hole concentrations (n and p) at an arbitrary temperature in **intrinsic** semiconductors?

Consider conduction band and valence band separately, and for each band:

$$n = \int_{\text{band}} f(E) g(E) dE$$

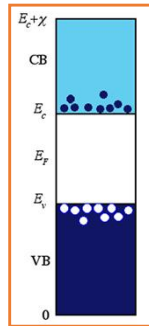
Define $g_c(E)$ and $g_v(E)$ as density of states in the conduction and valence band, respectively.



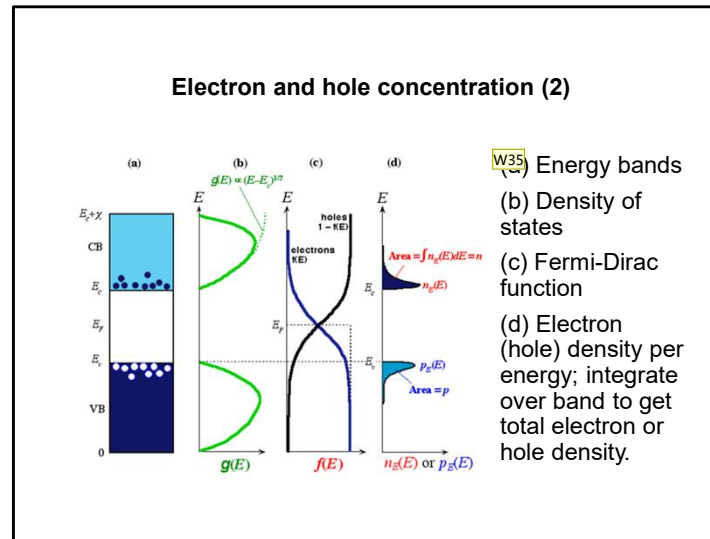
W34

This diagram explains carrier (electron and hole) distribution in a semiconductor using energy bands, density of states, and Fermi - Dirac statistics

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This diagram explains carrier (electron and hole) distribution in a semiconductor using energy bands, density of states, and Fermi - Dirac statistics.



Note E_c and E_v

work through the FD limits. E_F is $\frac{1}{2}$ ~mid-gap. If increase temperature $f(E)$ smudges and hence more electrons appear in CB. DOS zero at band edge.

W35

This diagram explains carrier (electron and hole) distribution in a semiconductor using energy bands, density of states, and Fermi - Dirac statistics

Panel (a): Energy bands (CB = Conduction Band, VB = Valence Band).

E_c : Conduction Band edge;

E_v : Valence Band edge.

E_F : Fermi Level (separates electron/hole regions).

$E_g = E_c - E_v$: Band gap.

Panel (b): Density of States ($g(E)$):

Number of available energy levels per unit energy.

For CB: $g(E) \propto (E - E_c)^{1/2}$

(green curve, rises with energy).

For VB: $g(E) \propto (E_v - E)^{1/2}$

(green curve, falls with energy).

Panel (c): Fermi - Dirac Distribution ($f(E)$):

Probability an energy level is occupied by an electron.

Electrons (blue): Follow Fermi - Dirac; dominant in CB.

Holes (black): Probability of unoccupation in VB ($1 - f(E)$).

Panel (d): Carrier Concentration:

Electrons (n): Integral of

$n_g(E) = g(E) \cdot f(E)$

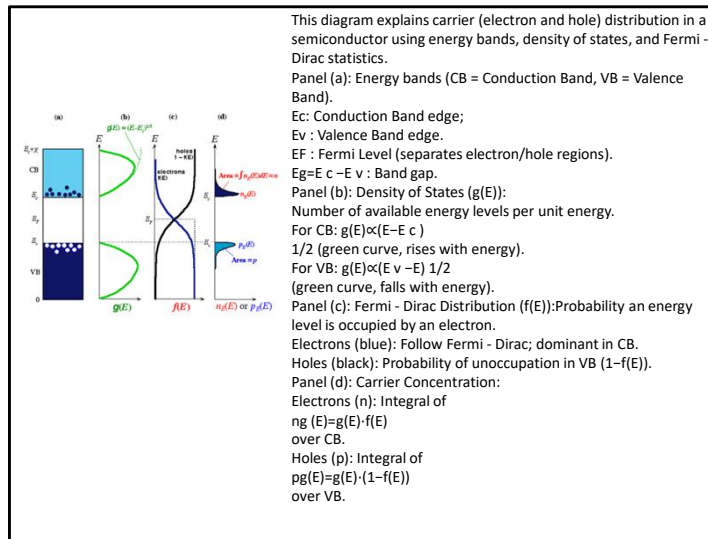
over CB (blue area).

Holes (p): Integral of

$p_g(E) = g(E) \cdot (1 - f(E))$

over VB (red area).

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Electron and hole concentration (3)

We can make some simplifications:

1. Assume energies in the conduction band are well above the Fermi energy ($E - E_F \gg k_b T$), so the Fermi-Dirac function is approximated by Boltzmann function:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_b T}\right)} \approx \exp\left(-\frac{E - E_F}{k_b T}\right)$$

Electron and hole concentration (4)

2. Assume that all electrons are close to the band bottom, we can use the following density of states:

$$g_c(E) = 8\pi\sqrt{2} \left(\frac{m_e^*}{h^2} \right)^{3/2} (E - E_c)^{1/2}$$

Also, the Fermi-Dirac function goes rapidly to zero as E increases, the upper limit of the integral can be made to be ∞ , which makes integral easier:

$$n = \int_{\text{conduction band}} f(E)g(E) dE \approx \int_{E_c}^{\infty} f(E)g(E) dE$$

Electron and hole concentration (5)

Putting all this together, and making the simplifications:

$$n \approx 8\pi\sqrt{2} \left(\frac{m_e^*}{h^2}\right)^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} \exp\left[-\frac{(E - E_F)}{k_b T}\right] dE$$

which leads to (*you don't need to do the integral!*):

$$n = 2 \left(\frac{2\pi m_e^* k_b T}{h^2}\right)^{3/2} \exp\left[-\frac{(E_c - E_F)}{k_b T}\right] = G_c(T) \exp\left[-\frac{(E_c - E_F)}{k_b T}\right]$$

Here, the effective density of states at the band edge is:

$$G_c(T) = 2 \left(\frac{2\pi m_e^* k_b T}{h^2}\right)^{3/2} \text{ [NB Kasap writes } N_c(T)\text{]}$$

Note that m_e^* is the effective mass of electrons in the CB.

Electron and hole concentration (6)

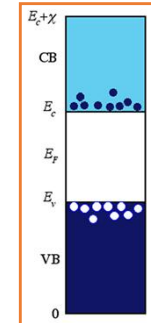
$$n = G_c(T) \exp \left[-\frac{(E_c - E_F)}{k_b T} \right]$$

$$G_c(T) = 2 \left(\frac{2\pi m_e^* k_b T}{h^2} \right)^{3/2}$$

n increases, as T increases, as we would expect. n is related to the energy difference between E_F and E_c .

If somehow the CB moved up in energy (i.e., E_c increased), n would decrease, again as we would expect.

E_F can be thought to be a material parameter related to the electron (or hole) density.



Electron and hole concentration (7)

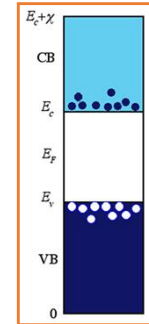
A similar analysis can be applied to the holes (see Kasap p330):

$$p = G_v(T) \exp \left[-\frac{(E_F - E_v)}{k_b T} \right]$$

$$G_v(T) = 2 \left(\frac{2\pi m_h^* k_b T}{h^2} \right)^{3/2}$$

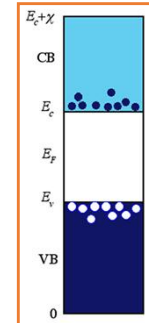
p is the number of holes in the valence band (per unit volume).

p is related to the energy difference between E_F and E_v .



Electron and hole concentration (8)

So we have derived expressions for n and p , with the only assumption being that the Fermi energy is somewhere in the bandgap, several $k_b T$ away from the conduction and valence band edges.



Law of mass action

There is a useful relation between n and p . Consider the product np :

$$\begin{aligned} np &= G_c(T)G_v(T) \exp\left[-\frac{(E_c-E_F)}{k_bT}\right] \exp\left[-\frac{(E_F-E_v)}{k_bT}\right] \\ &= G_c(T)G_v(T) \exp\left[-\frac{(E_c-E_v)}{k_bT}\right] \\ np &= G_c(T)G_v(T) \exp\left(-\frac{E_g}{k_bT}\right) \end{aligned}$$

This equation is very important. The product of the electron and hole concentration depends only on temperature, and on parameters specific to the semiconductor (e.g., E_g , m^*).

Law of mass action (2)

$$np = G_c(T)G_v(T) \exp\left(-\frac{E_g}{k_b T}\right)$$

This is the 'Law of Mass Action'. This equation holds for both **intrinsic** and **extrinsic** semiconductors!

For intrinsic $n = p$; so $np = p^2 = n^2 = ni^2$

[The sample has to be in thermal equilibrium, and not undergoing photoexcitation, for example].

If we introduce extra electrons into the semiconductor (e.g., by n -doping an intrinsic crystal), the hole concentration must decrease to keep np constant.

We can use this analysis to establish the Fermi energy position...

Fermi energy (intrinsic case)

An intrinsic semiconductor has equal number of electrons and holes (they are generated in pairs);
 $n = p = n_i$.

$$p = G_v(T) \exp\left[-\frac{(E_F - E_v)}{k_b T}\right]$$

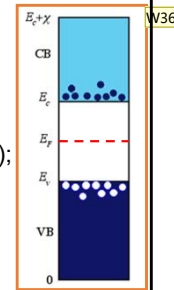
$$\text{and } p = \sqrt{np} = \sqrt{G_c G_v} \exp\left(-\frac{E_g}{2k_b T}\right)$$

Equate these & simplify: $E_F = E_v + \frac{1}{2}E_g - \frac{1}{2}k_b T \ln(G_c/G_v)$

Substitute G_c and G_v : $E_F = E_v + \frac{1}{2}E_g - \frac{3}{4}k_b T \ln(m_e^*/m_h^*)$

And so if $m_e^* \approx m_h^*$:

$$E_F = E_v + \frac{1}{2}E_g \quad (\text{Intrinsic Fermi energy is mid-gap})$$



W36

This diagram illustrates the energy band structure of a semiconductor, showing the distribution of electrons and holes in the conduction and valence bands.

CB (Conduction Band, light blue): Region where free electrons can move (conduct electricity).

VB (Valence Band, dark blue): Region where electrons are bound to atoms (no conduction, unless they jump to CB).

E_c

(Conduction Band Edge): Lowest energy level of the CB.

E_v

(Valence Band Edge): Highest energy level of the VB.

E_F (Fermi Level, dashed red line): Energy level where the probability of electron occupancy is 0.5 (at thermal equilibrium).

Blue dots: Free electrons in the CB.

White circles: Holes (empty spots) in the VB (act as positive charges).

Fermi energy (extrinsic, *n*-type)

An *n*-type doped semiconductor has $n > p$ (excess electrons over holes), but the product $np = n_i^2$ is unaltered.

Fermi energy is closer to the CB than the VB.

$$n = G_c(T) \exp \left[-\frac{(E_c - E_F)}{k_b T} \right]$$

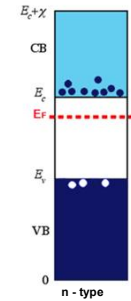
$(E_c - E_F)$ is smaller, making n larger.

$$p = G_v(T) \exp \left[-\frac{(E_F - E_v)}{k_b T} \right]$$

$(E_F - E_v)$ is larger, making p smaller.

But the product np is not affected by E_F .

$$np = G_c G_v \exp \left[-\frac{(E_c - E_v)}{k_b T} \right]$$



Fermi energy (extrinsic, *p*-type)

A *p*-type doped semiconductor has $p > n$ (excess holes), but the product $np = n_i^2$ is unaltered.

Fermi energy is closer to the VB than the CB.

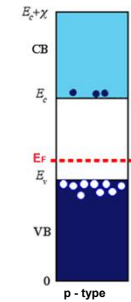
$$n = G_c(T) \exp\left[-\frac{(E_c - E_F)}{k_b T}\right]$$

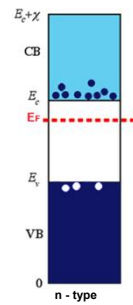
$(E_c - E_F)$ is larger, making n smaller.

$$p = G_v(T) \exp\left[-\frac{(E_F - E_v)}{k_b T}\right]$$

$(E_F - E_v)$ is smaller, making p larger.

But as before, the product np is not affected by the position of E_F .





This diagram illustrates the energy band structure of a semiconductor, showing the distribution of electrons and holes in the conduction and valence bands.

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E_c (Conduction Band Edge): Lowest energy level of the CB.

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Blue dots: Free electrons in the CB.

White circles: Holes (empty spots) in the VB (act as positive charges).

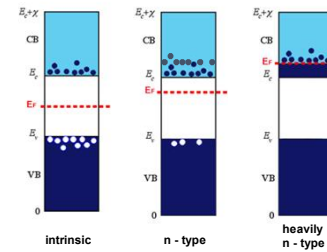
W37

Fermi energy as a function of doping

E_F is related to the number of electrons available for conduction.

In a metal, a large E_F implies a large number of conduction electrons.

In a semiconductor, the larger the E_F , the more electrons will be thermally excited into the CB at a given T .



W38

W37

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This diagram illustrates the energy band structures of three types of semiconductors:

Intrinsic semiconductor: It is a completely pure semiconductor with a crystalline structure and no impurities. At thermal equilibrium, the Fermi level (E_F) lies midway between the conduction band minimum (E_c) and the valence band maximum (E_v). Electrons in the valence band can gain enough energy to jump to the conduction band, creating an equal number of electron - hole pairs. The number of electrons in the conduction band is equal to the number of holes in the valence band, resulting in relatively weak electrical conductivity.

n - type semiconductor: Formed by doping an intrinsic semiconductor with pentavalent impurities (such as phosphorus or arsenic). These impurities provide additional electrons (act as electron donors). The Fermi level (E_F) shifts closer to the conduction band minimum (E_c). There are significantly more electrons in the conduction band than holes in the valence band. Electrons become the majority carriers, and holes become the minority carriers, so n - type semiconductors mainly conduct electricity through electrons.

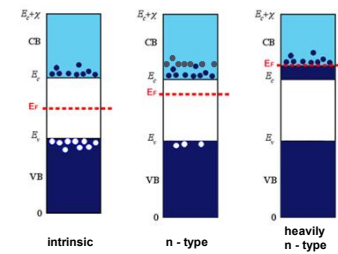
Heavily n - type semiconductor: A special case of an n - type semiconductor with a very high concentration of pentavalent impurities. The Fermi level (E_F) moves even closer to or even into the conduction band minimum (E_c). This leads to a large number of electrons in the conduction band, greatly enhancing the electrical conductivity.

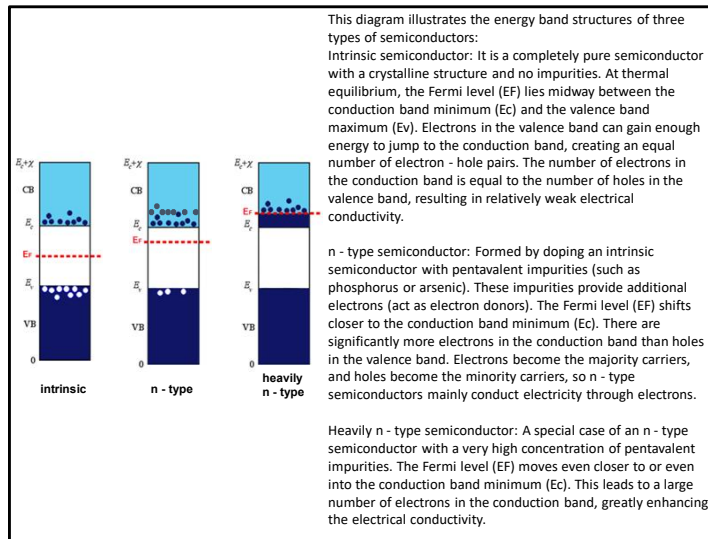
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Fermi energy as a function of doping (2)

If you dope a semiconductor very heavily, E_F can cross into a band.

In this example, E_F lies in the conduction band and the crystal has now become a **metal**. (degenerate)





Fermi energy – top tips

The Fermi Energy, E_F , is a material parameter, related to the number of electrons available for conduction.

In a metal, E_F lies inside an energy band.

In a semiconductor, E_F lies in the bandgap; close to the CB for n -type material, close to the VB for p -type material, and in the centre for intrinsic material.

A material in thermal equilibrium, with no external excitation (i.e. no connection to battery, or illumination) has a uniform E_F ($\Delta E_F = 0$).

Summary

Semiconductors are characterized by a band gap – a region of energy in which electrons are forbidden – separating the conduction and valence bands.

A metal has a partly filled band.

Conduction takes place by electrons (or holes) closest to the Fermi energy.

Intrinsic semiconductors have equal numbers of electrons and holes $n = p$.



W39



$$n = G_c(T) \exp \left[-\frac{(E_c - E_F)}{k_b T} \right]$$

$$p = G_v(T) \exp \left[-\frac{(E_F - E_v)}{k_b T} \right]$$

W39

This diagram shows the energy level structure of a material (likely a semiconductor or insulator), contrasting a filled/empty band configuration with a band - gap - induced separation.

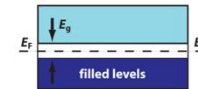
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Summary (2)

Extrinsic semiconductors incorporate dopant atoms which add extra electrons or holes.

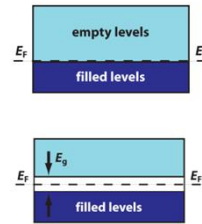


The position of the Fermi energy is related to the local carrier density, and at equilibrium, is constant across a device.



$$n = G_c(T) \exp \left[-\frac{(E_c - E_F)}{k_b T} \right]$$

$$p = G_v(T) \exp \left[-\frac{(E_F - E_v)}{k_b T} \right]$$



This diagram shows the energy level structure of a material (likely a semiconductor or insulator), contrasting a filled/empty band configuration with a band - gap - induced separation.